

House Price Prediction

Week 1 Internship Project — Summary of Findings

Dataset: Housing Prices Dataset (Kaggle) | Tools: Python, Pandas, Scikit-learn, Matplotlib, Seaborn

1. Key Features Influencing House Price

The correlation heatmap revealed that the most influential features on house price are area (correlation: 0.54), number of bathrooms (0.52), air conditioning (0.45), and location in a preferred area (0.33). This indicates that buyers place the highest value on spacious homes with modern amenities in well-located neighbourhoods. Features such as hot water heating and guest rooms showed comparatively weaker correlations with price.

2. Model Performance

Two regression models were trained and evaluated on an 80/20 train-test split:

Model	MAE	RMSE	R ² Score
Linear Regression	₹9,70,043	₹13,24,506	0.653
Random Forest	₹10,21,546	₹14,00,565	0.612

Linear Regression ($R^2 = 0.65$) slightly outperformed Random Forest ($R^2 = 0.61$), meaning the model can explain approximately 65% of the variation in house prices. The average prediction error was around ₹9,70,000 per property.

3. Surprising Finding

It was unexpected that the simpler Linear Regression model outperformed Random Forest. Typically, Random Forest handles complex patterns better. However, with only 545 records in this dataset, and the relationships between features and price being largely linear, the simpler model proved more effective. This is a useful reminder that more complex models do not always yield better results on small datasets.

4. Business Recommendation

Real estate businesses should prioritise area and number of bathrooms as the primary pricing factors, as these two features have the strongest influence on house value. Properties with air conditioning and those located in preferred areas consistently command a price premium and should be highlighted in listings. Investing in larger properties with modern amenities is likely to yield the highest returns.